Package ‘dilutionrisk’

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Type Package

Title Risk Assessment and Probability Estimation for Dilution Testing in Microbiology

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Description Provides comprehensive probability estimations and graphical displays for modelling and assessment of risk based on aerobic plate count (APC) on diluted testing. The package implements bounded distributions including truncated Poisson and truncated Poisson lognormal distributions for homogeneous and heterogeneous scenarios respectively. It includes functions for:
\itemize{
\item Probability of Detection (PD): Calculate detection probability for microbial contamination exceeding specification limits
\item Probability of Acceptance (PA): Assess sampling plan performance with binomial acceptance probabilities
\item Operating Characteristic (OC) Curves: Visual comparison of different dilution schemes
\item Coefficient of Variation (CV): Variability estimation for dilution schemes
\item True Concentration Estimation: Estimate true microbial concentration from diluted samples
\item Validation Tools: Compare theoretical and simulation results
}
The package supports both theoretical calculations and simulation-based validation approaches for robust risk assessment in food safety microbiology. It is particularly useful for designing and evaluating sampling plans for microbiological monitoring of foods.

License GPL (>= 2)

URL https://github.com/Mayooran1987/dilutionrisk

BugReports https://github.com/Mayooran1987/dilutionrisk/issues

Depends R (>= 4.0.0)

Imports ggplot2 (>= 3.3.0),
ggthemes (>= 4.2.0),
reshape2 (>= 1.4.4),
Rcpp (>= 1.0.0),
stats (>= 4.0.0),
methods (>= 4.0.0),
utils (>= 4.0.0)

LinkingTo Rcpp

Suggests testthat (>= 3.0.0),
covr (>= 3.5.0),
knitr (>= 1.30),
rmarkdown (>= 2.7),
pkgdown (>= 2.0.0),
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Repository https://github.com/Mayooran1987/dilutionrisk

RemoteUsername Mayooran1987

RemoteRepo dilutionrisk

RemoteHost api.github.com

RemoteRef HEAD

Contents

`cv_heterogeneous` *Coefficient of Variation for Heterogeneous Batches*

Description

Estimates the coefficient of variation (CV) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
cv_heterogeneous(mu, sd, a, b, f, u, USL, n_sim)

cv_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)

cv_curves_heterogeneous(mu_low, mu_high, sd, a, b, f, u, USL, n_sim)
```

Arguments

`mu`	Mean microbial count on the log scale. Must satisfy `log(a) - log(f*u) <= mu <= log(b) - log(f*u)`.
`sd`	Standard deviation of the normal distribution on the log scale. Must be positive.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.
`u`	Amount placed on the plate. Must be positive.
`USL`	Upper specification limit for microbial count.
`n_sim`	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
`mu_low`	Lower bound of mean microbial count (μ) for x-axis in graphical displays (log scale).
`mu_high`	Upper bound of mean microbial count (μ) for x-axis in graphical displays (log scale).

Value

Numeric value representing the coefficient of variation for the heterogeneous batch.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.

Examples

```
## Not run:
# Basic usage
cv_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                 f = 0.01, u = 0.1, USL = 1000, n_sim = 50000)

# Multiple dilution schemes
cv_heterogeneous_multiple(mu = 2, sd = 0.2, a = 0, b = 300,
                          f = c(0.01, 0.1), u = c(0.1, 0.1),
                          USL = 1000, n_sim = 50000)

# Plot CV curves
cv_curves_heterogeneous(mu_low = -5, mu_high = 10, sd = 0.2,
                        a = 0, b = 300, f = c(0.01, 0.1),
                        u = c(0.1, 0.1), USL = 1000, n_sim = 50000)

## End(Not run)
```

---

cv_homogeneous — *Coefficient of Variation for Homogeneous Batches*

Description

Estimates the coefficient of variation (CV) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
cv_homogeneous(lambda, a, b, f, u, USL, n_sim)

cv_homogeneous_multiple(lambda, a, b, f, u, USL, n_sim)

cv_curves_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim)
```

Arguments

`lambda`	Expected microbial count (λ). Must be positive.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.

u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n_sim	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
lambda_low	Lower bound of expected microbial count for x-axis in graphical displays.
lambda_high	Upper bound of expected microbial count for x-axis in graphical displays.

Value

Numeric value representing the coefficient of variation for the homogeneous batch.

Examples

```
## Not run:
# Basic usage
cv_homogeneous(lambda = 2000, a = 0, b = 300,
               f = 0.01, u = 0.1, USL = 1000, n_sim = 50000)

# Multiple dilution schemes
cv_homogeneous_multiple(lambda = 2000, a = 0, b = 300,
                        f = c(0.01, 0.1), u = c(0.1, 0.1),
                        USL = 1000, n_sim = 50000)

# Plot CV curves
cv_curves_homogeneous(lambda_low = 1000, lambda_high = 8000,
                      a = 0, b = 300, f = c(0.01, 0.1),
                      u = c(0.1, 0.1), USL = 1000, n_sim = 50000)

## End(Not run)
```

OC_curves_heterogeneous
Operating Characteristic (OC) Curves for Heterogeneous Batches

---

Description

Generates operating characteristic curves comparing different dilution schemes when samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
OC_curves_heterogeneous(
  c,
  mu_low,
  mu_high,
  sd,
  a,
  b,
  f,
  u,
```

```
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

c	Acceptance number (maximum number of defective units allowed).
mu_low	Lower bound of mean microbial count for x-axis (log scale).
mu_high	Upper bound of mean microbial count for x-axis (log scale).
sd	Standard deviation of the normal distribution on the log scale.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing OC curves for different dilution schemes.

Examples

```
## Not run:
OC_curves_heterogeneous(c = 2, mu_low = 4, mu_high = 9, sd = 0.2,
                        a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                        USL = 1000, n = 5)

## End(Not run)
```

OC_curves_homogeneous *Operating Characteristic (OC) Curves for Homogeneous Batches*

Description

Generates operating characteristic curves comparing different dilution schemes when samples are collected from a homogeneous batch.

Usage

```
OC_curves_homogeneous(
  c,
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

`c`	Acceptance number (maximum number of defective units allowed).
`lambda_low`	Lower bound of expected microbial count for x-axis.
`lambda_high`	Upper bound of expected microbial count for x-axis.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Vector of final dilution factors.
`u`	Vector of amounts placed on the plate.
`USL`	Upper specification limit for microbial count.
`n`	Number of samples inspected. Must be positive integer.
`type`	Type of calculation: "theory" (default) or "simulation".
`n_sim`	Number of simulations. Required when `type = "simulation"`.

Details

Operating characteristic curves show the probability of acceptance as a function of the true microbial concentration. They are useful for comparing the performance of different dilution schemes and sampling plans.

Value

A ggplot object showing OC curves for different dilution schemes.

Examples

```
## Not run:
OC_curves_homogeneous(c = 2, lambda_low = 0, lambda_high = 5000,
                      a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                      USL = 1000, n = 5)

## End(Not run)
```

pd_curves_heterogeneous
Probability of Detection Curves for Heterogeneous Batches

Description

Generates probability of detection curves comparing different dilution schemes when samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
pd_curves_heterogeneous(
  mu_low,
  mu_high,
  sd,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

Arguments

mu_low	Lower bound of mean microbial count for x-axis (log scale).
mu_high	Upper bound of mean microbial count for x-axis (log scale).
sd	Standard deviation of the normal distribution on the log scale.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing PD curves for different dilution schemes.

Examples

```
## Not run:
pd_curves_heterogeneous(mu_low = 0, mu_high = 10, sd = 0.2,
                        a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                        USL = 1000)

## End(Not run)
```

pd_curves_homogeneous *Probability of Detection Curves for Homogeneous Batches*

Description

Generates probability of detection curves comparing different dilution schemes when samples are collected from a homogeneous batch.

Usage

```
pd_curves_homogeneous(
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

Arguments

lambda_low	Lower bound of expected microbial count for x-axis.
lambda_high	Upper bound of expected microbial count for x-axis.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing PD curves for different dilution schemes.

Examples

```
## Not run:
pd_curves_homogeneous(lambda_low = 0, lambda_high = 5000,
                      a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                      USL = 1000)

## End(Not run)
```

---

pd_validation_heterogeneous *Comparison based on probability of detection curves for different dilution schemes when diluted samples collected from a heterogeneous batch.*

Description

pd_validation_heterogeneous provides the probability of detection curves for validate the results when samples collected from a heterogeneous batch.

Usage

```
pd_validation_heterogeneous(mu_low,  mu_high, sd, a, b, f, u, USL, n_sim)
```

Arguments

mu_low	the lower value of the mean microbial count (μ) for use in the graphical display's x-axis (on the log scale).
mu_high	the upper value of the mean microbial count (μ) for use in the graphical display's x-axis (on the log scale).
sd	the standard deviation of the normal distribution (on the log scale).
a	lower domain of the number of cell counts.
b	upper domain of the number of cell counts.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n_sim	number of simulations (large simulations provide more precise estimations).

Details

pd_curves_heterogeneous provides probability of detection curves for different dilution schemes when samples collected from a heterogeneous batch (this section will be updated later on).

Value

Probability of detection curves when diluted samples collected from a heterogeneous batch.

Examples

```
mu_low <- 1
mu_high <- 12
sd <- 0.2
a <- 0
b <- 300
f <- 0.01
u <- 0.1
USL <- 1000
n_sim <- 50000
pd_validation_heterogeneous(mu_low,  mu_high, sd, a, b, f, u, USL, n_sim)
```

`pd_validation_homogeneous`
Comparison based on probability of detection curves for different dilution schemes when diluted samples collected from a homogeneous batch.

Description

`pd_validation_homogeneous` provides the probability of detection curves for validate the results when samples collected from a homogeneous batch.

Usage

```
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim)
```

Arguments

`lambda_low`	the lower value of the expected microbial count (λ) for use in the graphical display's x-axis.
`lambda_high`	the upper value of the expected microbial count (λ) for use in the graphical display's x-axis.
`a`	lower domain of the number of microbial count.
`b`	upper domain of the number of microbial count.
`f`	final dilution factor.
`u`	amount put on the plate.
`USL`	upper specification limit.
`n_sim`	number of simulations (large simulations provide more precise estimations).

Details

`pd_curves_homogeneous` provides probability of detection curves for different dilution schemes when samples collected from a homogeneous batch (this section will be updated later on).

Value

Probability of detection curves when diluted samples collected from a homogeneous batch.

Examples

```
lambda_low <- 0
lambda_high <- 5000
a <- 0
b <- 300
f <- 0.01
u <- 0.1
USL <- 1000
n_sim <- 50000
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim = 500)
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim = 50000)
```

prob_acceptance_heterogeneous
Probability of Acceptance for Heterogeneous Batches

Description

Calculates the probability of acceptance (PA) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
prob_acceptance_heterogeneous(
  c,
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)

prob_acceptance_heterogeneous_multiple(
  c,
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

c	Acceptance number (maximum number of defective units allowed).
mu	Mean microbial count on the log scale. Must satisfy `log(a) - log(f*u) <= mu <= log(b) - log(f*u)`.
sd	Standard deviation of the normal distribution on the log scale. Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than `a`.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.

USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

The probability of acceptance is calculated using the binomial distribution:

$$P_a = P(X \le c) = \sum_{k=0}^{c} \binom{n}{k} p_d^k (1 - p_d)^{n-k}$$

where p_d is the probability of detection and n is the number of samples inspected.

Value

Numeric value representing the probability of acceptance.

Examples

```
## Not run:
# Basic usage
prob_acceptance_heterogeneous(c = 2, mu = 7, sd = 0.2, a = 0, b = 300,
                              f = 0.01, u = 0.1, USL = 1000, n = 5)

# Multiple dilution schemes
prob_acceptance_heterogeneous_multiple(c = 2, mu = 7, sd = 0.2,
                                       a = 0, b = 300,
                                       f = c(0.01, 0.1), u = c(0.1, 0.1),
                                       USL = 1000, n = 5)

## End(Not run)
```

---

prob_acceptance_heterogeneous_multiple

Probability of acceptance estimation when diluted samples are collected from a heterogeneous batch.

Description

prob_acceptance_heterogeneous_multiple provides a probability of acceptance in the original sample when samples collected from a heterogeneous batch.

Usage

```
prob_acceptance_heterogeneous_multiple (c, mu, sd, a, b, f, u, USL, n, type, n_sim)
```

Arguments

`c`	acceptance number
`mu`	the mean microbial count (on the log scale).
`sd`	the standard deviation of the normal distribution (on the log scale).
`a`	lower domain of the number of cell counts.
`b`	upper domain of the number of cell counts.
`f`	final dilution factor.
`u`	amount put on the plate.
`USL`	upper specification limit.
`n`	number of samples which are used for inspection.
`type`	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
`n_sim`	number of simulations (large simulations provide more precise estimations).

Details

`prob_acceptance_heterogeneous_multiple` provides a probability of acceptance when diluted samples are collected from a heterogeneous batch (this section will be updated later on).

Value

Probability of acceptance when samples collected from a heterogeneous batch.

Examples

```
c <- 2
mu <- 7
sd <- 0.2
a <- 0
b <- 300
f <- c(0.01,0.1,1)
u <- c(0.1,0.1,0.1)
USL <- 1000
n <- 5
n_sim <- 50000
prob_acceptance_heterogeneous_multiple (c, mu, sd, a, b, f, u, USL, n)
```

---

prob_acceptance_homogeneous
Probability of Acceptance for Homogeneous Batches

Description

Calculates the probability of acceptance (PA) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
prob_acceptance_homogeneous(
  c,
  lambda,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)

prob_acceptance_homogeneous_multiple(
  c,
  lambda,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

`c`	Acceptance number (maximum number of defective units allowed).
`lambda`	Expected microbial count (λ). Must be positive.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.
`u`	Amount placed on the plate. Must be positive.
`USL`	Upper specification limit for microbial count.
`n`	Number of samples inspected. Must be positive integer.
`type`	Type of calculation: "theory" (default) or "simulation".
`n_sim`	Number of simulations. Required when `type = "simulation"`.

Details

The probability of acceptance is calculated using the binomial distribution:

$$P_a = P(X \leq c) = \sum_{k=0}^{c} \binom{n}{k} p_d^k (1 - p_d)^{n-k}$$

where p_d is the probability of detection and n is the number of samples inspected.

Value

Numeric value representing the probability of acceptance.

Examples

```
## Not run:
# Basic usage
prob_acceptance_homogeneous(c = 2, lambda = 2000, a = 0, b = 300,
                            f = 0.01, u = 0.1, USL = 1000, n = 5)

# Multiple dilution schemes
prob_acceptance_homogeneous_multiple(c = 2, lambda = 2000, a = 0, b = 300,
                                     f = c(0.01, 0.1), u = c(0.1, 0.1),
                                     USL = 1000, n = 5)

## End(Not run)
```

---

prob_acceptance_homogeneous_multiple

Probability of acceptance estimation for multiple dilution schemes when diluted samples are collected from a homogeneous batch.

Description

prob_acceptance_homogeneous_multiple provides a probability of acceptance for multiple dilution schemes in the original sample when samples collected from a homogeneous batch

Usage

```
prob_acceptance_homogeneous_multiple(c, lambda, a, b, f, u, USL, n, type, n_sim)
```

Arguments

c	acceptance number
lambda	the expected microbial count (λ).
a	lower domain of the number of microbial count.
b	upper domain of the number of microbial count.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n	number of samples which are used for inspection.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide more precise estimations).

Details

prob_detection_homogeneous_multiple provides a probability of acceptance for multiple dilution schemes in the original sample when samples collected from a homogeneous batch (this section will be updated later on).

Value

Probability of acceptance when diluted samples are collected from a homogeneous batch.

Examples

```
c <- 2
lambda <- 1000
a <- 0
b <- 300
f <- c(0.01,0.1,1)
u <- c(0.1,0.1,0.1)
USL <- 1000
n <- 5
n_sim <- 50000
prob_acceptance_homogeneous_multiple(c, lambda, a, b, f, u, USL, n)
```

---

prob_detection_heterogeneous

Probability of Detection for Heterogeneous Batches

Description

Calculates the probability of detection (PD) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
prob_detection_heterogeneous(
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)

prob_detection_heterogeneous_multiple(
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

Arguments

`mu`	Mean microbial count on the log scale. Must satisfy `log(a) - log(f*u) <= mu <= log(b) - log(f*u)`.
`sd`	Standard deviation of the normal distribution on the log scale. Must be positive.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.
`u`	Amount placed on the plate. Must be positive.
`USL`	Upper specification limit for microbial count.
`type`	Type of calculation: "theory" (default) or "simulation".
`n_sim`	Number of simulations. Required when `type = "simulation"`.

Details

The probability of detection is defined as the probability that the estimated microbial count exceeds the upper specification limit (USL). For heterogeneous batches, the count follows a truncated Poisson-lognormal distribution.

The theoretical probability is calculated by integrating over the lognormal mixing distribution:

$$P_d = 1 - \sum_{i=1}^{USL \times f \times u} \frac{1}{i!\sqrt{2\pi}\sigma} \int_0^\infty \frac{t^{i-1} \exp(-0.5((\log(t) - \mu_d)/\sigma)^2)}{e^t - 1} dt$$

Value

Numeric value representing the probability of detection.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.

Examples

```
## Not run:
# Theoretical calculation
prob_detection_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                             f = 0.01, u = 0.1, USL = 1000)

# Simulation-based calculation
prob_detection_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                             f = 0.01, u = 0.1, USL = 1000,
                             type = "simulation", n_sim = 50000)

## End(Not run)
```

---

prob_detection_heterogeneous_multiple

Probability of detection estimation for multiple dilution schemes when diluted samples are collected from a heterogeneous batch.

Description

prob_detection_heterogeneous_multiple provides a probability of detection for multiple dilution schemes in the original sample when samples collected from a heterogeneous batch.

Usage

```
prob_detection_heterogeneous_multiple(mu, sd, a, b, f, u, USL, type , n_sim)
```

Arguments

mu	the mean microbial count (on the log scale).
sd	the standard deviation of the normal distribution (on the log scale).
a	lower domain of the number of cell counts.
b	upper domain of the number of cell counts.
f	vector of final dilution factor.
u	vector of amount put on the plate.
USL	upper specification limit.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide a more precise estimation).

Details

prob_detection_heterogeneous_multiple provides a probability of detection when diluted samples are collected from a heterogeneous batch. We define the random variable X_i is the number of colonies on the i^{th} plate. In practice, the acceptance for countable numbers of colonies on a plate must be between 30 and 300. Therefore, we can utilise bounded distributions to model the number of colonies on a plate. In the heterogeneous case, we employed truncated Poisson lognormal distribution to the model. (this section will be updated later on).

Value

Probability of detection when samples collected from a heterogeneous batch.

Examples

```
mu <- 7
sd <- 0.2
a <- 0
b <- 300
f <- c(0.01,0.1,1)
u <- c(0.1,0.1,0.1)
USL <- 1000
n_sim <- 50000
prob_detection_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)
```

prob_detection_homogeneous
Probability of Detection for Homogeneous Batches

Description

Calculates the probability of detection (PD) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
prob_detection_homogeneous(
  lambda,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)

prob_detection_homogeneous_multiple(
  lambda,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

Arguments

lambda	Expected microbial count (λ). Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

The probability of detection is defined as the probability that the estimated microbial count exceeds the upper specification limit (USL). For homogeneous batches, the count follows a truncated Poisson distribution.

The theoretical probability is calculated as:

$$P_d = 1 - \sum_{i=1}^{USL \times f \times u} \frac{(\lambda f u)^i}{i!(e^{\lambda f u} - 1)}$$

Value

Numeric value representing the probability of detection.

References

- Schothorst, M. van, Zwietering, M.H., Ross, T., Buchanan, R.L., Cole, M.B., 2009. Relating microbiological criteria to food safety objectives and performance objectives. Food Control 20, 967–979.

Examples

```
## Not run:
# Theoretical calculation
prob_detection_homogeneous(lambda = 2000, a = 0, b = 300,
                           f = 0.01, u = 0.1, USL = 1000)

# Simulation-based calculation
prob_detection_homogeneous(lambda = 2000, a = 0, b = 300,
                           f = 0.01, u = 0.1, USL = 1000,
                           type = "simulation", n_sim = 50000)

## End(Not run)
```

`prob_detection_homogeneous_multiple`

Probability of detection estimation for multiple dilution schemes when diluted samples are collected from a homogeneous batch.

Description

`prob_detection_homogeneous_multiple` provides a probability of detection for multiple dilution schemes in the original sample when samples collected from a homogeneous batch.

Usage

```
prob_detection_homogeneous_multiple(lambda, a, b, f, u, USL, type, n_sim)
```

Arguments

`lambda`	the expected microbial count (λ).
`a`	lower domain of the number of microbial count.
`b`	upper domain of the number of microbial count.
`f`	final dilution factor.
`u`	amount put on the plate.

USL	upper specification limit.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide more precise estimations).

Details

prob_detection_homogeneous_multiple provides a probability of detection when the diluted sample has homogeneous contaminants. We define the random variable X_i is the number of colonies on the i^{th} plate. In practice, the acceptance for countable numbers of colonies on a plate must be between 30 and 300. Therefore, we can utilise bounded distributions to model the number of colonies on a plate. In the homogeneous case, we employed truncated Poisson distribution to model (this section will be updated later on).

Value

Probability of detection when diluted samples are collected from a homogeneous batch.

Examples

```
lambda <- 1000
a <- 0
b <- 300
f <- c(0.01,0.1,1)
u <- c(0.1,0.1,0.1)
USL <- 1000
n_sim <- 50000
prob_detection_homogeneous_multiple(lambda, a, b, f, u, USL)
```

---

rtrunpoilog — *Truncated Poisson-Lognormal Random Number Generation*

Description

Generates random deviates from a truncated Poisson-lognormal distribution.

Usage

```
rtrunpoilog(n, mu, sd, a, b)
```

Arguments

n	Number of observations. If length(n) > 1, the length is taken to be the number required.
mu	Mean of the lognormal distribution on the log scale.
sd	Standard deviation of the lognormal distribution on the log scale.
a	Lower truncation point (lower bound of cell count).
b	Upper truncation point (upper bound of cell count).

Details

The Poisson-lognormal distribution is a mixture distribution where:

- $\Lambda \sim \text{Lognormal}(\mu, \sigma^2)$
- $X|\Lambda = \lambda \sim \text{Poisson}(\lambda)$

The truncation limits the support to $a \leq X \leq b$.

Value

A vector of random numbers from the truncated Poisson-lognormal distribution.

Examples

```
## Not run:
# Generate 100 random values
set.seed(123)
rtrunpoilog(n = 100, mu = 0, sd = 1, a = 0, b = 300)

# Generate a single value
rtrunpoilog(n = 1, mu = 0.5, sd = 0.8, a = 10, b = 200)

## End(Not run)
```

`true_concentration_heterogeneous`
True Concentration Estimation for Heterogeneous Batches

Description

Estimates the true microbial concentration in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
true_concentration_heterogeneous(mu, sd, a, b, f, u, USL, n_sim)

true_concentration_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)

true_concentration_curves_heterogeneous(
  mu_low,
  mu_high,
  sd,
  a,
  b,
  f,
  u,
  USL,
  n_sim
)
```

Arguments

`mu`	Mean microbial count on the log scale. Must satisfy `log(a) - log(f*u) <= mu <= log(b) - log(f*u)`.
`sd`	Standard deviation of the normal distribution on the log scale.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.
`u`	Amount placed on the plate. Must be positive.
`USL`	Upper specification limit for microbial count.
`n_sim`	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
`mu_low`	Lower bound of mean microbial count for x-axis (log scale).
`mu_high`	Upper bound of mean microbial count for x-axis (log scale).

Details

The true concentration is estimated as:

$$C = \frac{X}{f \times u}$$

where X is the count of microorganisms on a plate following a truncated Poisson-lognormal distribution, f is the final dilution factor, and u is the amount placed on the plate.

Value

Numeric value representing the estimated true concentration.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.

Examples

```
## Not run:
# Basic usage
true_concentration_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                                 f = 0.01, u = 0.1, USL = 1000, n_sim = 50000)

# Multiple dilution schemes
true_concentration_heterogeneous_multiple(mu = 2, sd = 0.2, a = 0, b = 300,
                                          f = c(0.01, 0.1), u = c(0.1, 0.1),
                                          USL = 1000, n_sim = 50000)

# Plot concentration curves
true_concentration_curves_heterogeneous(mu_low = 0, mu_high = 10, sd = 0.2,
                                        a = 0, b = 300, f = c(0.01, 0.1),
                                        u = c(0.1, 0.1), USL = 1000,
                                        n_sim = 50000)

## End(Not run)
```

`true_concentration_homogeneous`
True Concentration Estimation for Homogeneous Batches

Description

Estimates the true microbial concentration in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
true_concentration_homogeneous(lambda, a, b, f, u, USL, n_sim)

true_concentration_homogeneous_multiple(lambda, a, b, f, u, USL, n_sim)

true_concentration_curves_homogeneous(
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  n_sim
)
```

Arguments

`lambda`	Expected microbial count (λ). Must be positive.
`a`	Lower bound of cell count domain. Must be non-negative.
`b`	Upper bound of cell count domain. Must be greater than `a`.
`f`	Final dilution factor. Must be between 0 and 1.
`u`	Amount placed on the plate. Must be positive.
`USL`	Upper specification limit for microbial count.
`n_sim`	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
`lambda_low`	Lower bound of expected microbial count for x-axis.
`lambda_high`	Upper bound of expected microbial count for x-axis.

Details

The true concentration is estimated as:

$$C = \frac{X}{f \times u}$$

where X is the count of microorganisms on a plate following a truncated Poisson distribution, f is the final dilution factor, and u is the amount placed on the plate.

Value

Numeric value representing the estimated true concentration.

Examples

```
## Not run:
# Basic usage
true_concentration_homogeneous(lambda = 2000, a = 0, b = 300,
                               f = 0.01, u = 0.1, USL = 1000, n_sim = 50000)

# Multiple dilution schemes
true_concentration_homogeneous_multiple(lambda = 2000, a = 0, b = 300,
                                        f = c(0.01, 0.1), u = c(0.1, 0.1),
                                        USL = 1000, n_sim = 50000)

# Plot concentration curves
true_concentration_curves_homogeneous(lambda_low = 0, lambda_high = 5000,
                                      a = 0, b = 300, f = c(0.01, 0.1),
                                      u = c(0.1, 0.1), USL = 1000,
                                      n_sim = 50000)

## End(Not run)
```

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